

	shape	[1]
9(a)(i)	At constant pressure, $V \propto T$ $\frac{V_1}{V_2} = \frac{T_1}{T_2}$ $\frac{500}{V_2} = \frac{250}{180}$ $V_2 = 360 \text{ cm}^3$ work done on gas $= -p\Delta V$ $= -(80 \times 10^3)(360 - 500) \times 10^{-6}$ $= 11.2 \text{ J}$	[1] for correct V_2 [1] for substitution [1] for answer
9(a)(ii)	Using $pV=nRT$, $n = \frac{pV}{RT} = \frac{(80 \times 10^3)(500 \times 10^{-6})}{(8.31)(250)} = 0.01925 \text{ mol}$ Change in internal energy, $\Delta U = \frac{3}{2} nR\Delta T = \frac{3}{2} (0.01925)(8.31)(180 - 250) = -16.8 \text{ J}$ OR Change in internal energy, $\Delta U = \frac{3}{2} nR\Delta T = \frac{3}{2} p\Delta V$ $= \frac{3}{2} (80 \times 10^3)(360 - 500) \times 10^{-6} = -16.8 \text{ J}$	[1] for substitution [1] for answer
9(a)(iii)	Using first law of thermodynamics,	[1] for answer (must be positive)

	$\Delta U = Q + W$ $Q = \Delta U - W$ $= -16.8 - 11.2$ $= -28.0 \text{ J}$ Hence, the amount of heat lost is 28.0 J.	
9(b)(i)	Using $pV=nRT$, $p = \frac{nRT}{V} = \frac{(0.01925)(8.31)(250)}{(360 \times 10^{-6})} = 111 \text{ kPa}$	[1] for substitution [1] for answer
9(b)(ii)	Change in internal energy, $\Delta U = \frac{3}{2} nR\Delta T = \frac{3}{2} (0.01925)(8.31)(250 - 180) = 16.8 \text{ J}$ Since there is no work done on the gas as volume is constant, $\Delta U = Q$ $Q = mc\Delta T = 16.8$ $c = \frac{16.8}{(0.23 \times 10^{-3})(250 - 180)} = 1043 \approx 1040 \text{ J kg}^{-1} \text{ K}^{-1}$	[1] correct explanation and application of $\Delta U = Q$ [1] for substitution [1] for answer
9(b)(iii)	$\frac{3}{2} nRT = \frac{1}{2} m_{\text{total}} \langle c^2 \rangle$ $c_{\text{r.m.s.}} = \sqrt{\frac{3nRT}{m}} = \sqrt{\frac{3(0.01925)(8.31)(250)}{(0.23 \times 10^{-3})}}$ $= 722.2 \approx 720 \text{ m s}^{-1}$	[1] substitution [1] answer
9(b)(iv)	The root-mean-square speed of the particles would remain the same. This is because the root-mean-square speed for each particle is only dependent on the temperature and not on the amount of gas, given that the mass of each particle is the same.	[1] [1]
9(c)	pressure / kPa 111 80 0 0 360 500 volume / cm³ (b) (c) (a)	[1] for each process [-1] for missing axis labels

9(d)	Since the work done by the gas during expansion is greater than the work done on the gas during compression, there is a <u>net work done by the gas</u> in each cycle.	[1]
	Since there is no change in internal energy in each cycle, the <u>gas gains heat</u> in each cycle.	[1]